Assignment 1: Full Stack Web Development

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1. Question 1

Original

Full Stack Development refers to the process of developing both the front-end (user interface) and the back-end (server-side logic and database) of a web application.

Correct

Full Stack Development is the process of developing both the **frontend (user interface)** and the **backend (server-side logic and database)** of a web application. A **Full Stack Developer** can work on every layer of an application, including the frontend, backend, and database.

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2. Question 2

Correct

The **frontend** is the part of a website that users interact with directly. It includes everything users see, click, and type into on their browser.

Technologies: HTML, CSS, JavaScript, React, Angular, Vue.js.

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3. Question 3

Correct

The **backend** is the server-side part of a website. It handles business logic, user authentication, API requests, and communication with the database.

Technologies: Node.js, Java, Python, PHP, .NET, etc.

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4. Question 4

Better wording

A **database** is an organized collection of data stored electronically. It stores information permanently so it can be retrieved, updated, or deleted whenever required.

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5. Question 5

Change this:

192.168.1.1

to something like

142.250.190.78

because **192.168.1.1** is a **private IP address**, not a public website IP.

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6. Question 6

Correct

An **IP (Internet Protocol) address** is a unique numerical address assigned to every device connected to a network. It helps devices identify and communicate with each other over the internet.

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7. Question 7

Very good. No changes needed.

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8. Question 8

Better wording

An **API (Application Programming Interface)** allows communication between the frontend and backend. It receives requests from the frontend, processes them through the backend, and sends responses back.

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9. Question 9

Perfect.

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10. Question 10

Better wording

Developer Tools (DevTools) are built-in browser tools used to inspect, debug, and test websites.

Architecture Diagram

Instead of

Queries Data

write

Reads/Writes Data

because the backend both **reads** and **writes** data.

Website Working Flow

Change

Server → Formats data into HTML/JSON Response

to

Server → Processes the request and sends an HTML or JSON response.

Student Registration System

Instead of

hashes the password

write

hashes the password before storing it in the database.

Amazon Example

Very good.

Website Observation

Instead of

I saw a few yellow warning messages

write

I observed some warning messages in the Console tab, but there were no critical errors.

This sounds more professional.

Reflection 16

Instead of

browser first contacts a DNS server

write

the browser first sends a DNS request to translate the domain name into an IP address.

Reflection 17

Excellent.

Just change

It remembers everything

to

The database stores all user information permanently, allowing users to access their accounts whenever they log in.

One Technical Correction

You wrote:

Frontend → Backend → Database

Actually, the flow is:

User

↓

Browser

↓

DNS

↓

Server (Backend)

↓

Database

↓

Server

↓

Browser

↓

User

The **browser loads the frontend (HTML, CSS, and JavaScript)** after receiving it from the server. The frontend then communicates with the backend using APIs for dynamic data.